ATOM COUNT CARD

QUEEN BAKING SODA

A. PRODUCT IDENTITY





- **Product:** Baking soda, net weight 125g
- **Compound name and formula:** Sodium Bicarbonate, $NaHCO_3$
- **Why it matters:** in manufacturing PCB (Printed Circuit Board) the sodium carbonate acts as abrasive material in cleaning process that removes residue from small holes in circuit boards.
- **Student:** Mary France Angel C. Serrano, Chemistry for Engineers

B. BASE QUANTITATIVE DATA

QUANTITY	VALUE	SOURCE	Precision/Sig. figs.
Net weight	125g	Printed on the product box	3 s.f.
Serving size	2.4g	Printed on the nutrition label	2 s.f.
Servings per container	52 servings	Printed on the product box	2 s.f.
Chemical Formula	$NaHCO_3$	PubChem	Exact count
Atomic masses	Na = 22.990, H = 1.008, C = 12.011, O = 15.999 g/mol	CIAAW Abridged Standard Atomic Weights (2024)	Not limiting
Avogadro constant	$6.02214076 \times 10^{23}\ mol^{-1}$	BIPM, International System of Units (SI)	Exact

C. MOLAR-MASS BUILD-UP

Element	Atoms	Atomic mass (g/mol)	(g/mol)	%by mass
Na	1	22.990	22.990	27.37%
H	1	1.008	1.008	1.20%
C	1	12.011	12.011	14.30%
O	3	15.999	47.997	57.14%
$NaHCO_3$	6		84.006	100.01%

D. THE MOLE MAP: one 125g box

Mass: 125 g ($NaHCO_3$) → ÷ 84.006 g/mol: 125g ÷ 84.006 g/mol = 1.487988953 → Mole: 1.488 mol ($NaHCO_3$) → × $6.02214076 \times 10^{23}$: 1.488 mol × $6.02214076 \times 10^{23}$ = 8.960878925 → Particle: 8.96×10^{23} Formula units

The calculated mol was rounded off into 3 decimal place.

E. MOLE - CONVERSION CALCULATION

1 Molar mass of $NaHCO_3$

$M = 1(22.990) + 1(1.008) + 1(12.011) + 3(15.999)$

$= 22.990 + 1.008 + 12.011 + 47.997 =$ **86.006 g/mol**

2 Label bridge: package → mass

$$1\ \text{box} \times \frac{125\text{g}}{1\ \text{box}} = \textbf{125g baking soda}$$

3 Mass → moles

$$125\text{g} \times \frac{1\ \text{mol}}{84.006\text{g}} = 1.4880\ \text{mol} \rightarrow \textbf{1.5 mol baking soda}$$

4 Moles → molecules

$$1.4880\ \cancel{\text{mol}} \times \frac{6.02214076 \times 10^{23}\ \text{molecules}}{1\ \cancel{\text{mol}}} = 8.9609 \times 10^{24}$$

→ **8.96×10^{24} molecules**

5 Atoms of one element inside the formula

$$8.9609 \times 10^{24}\ \cancel{\text{molecules}} \times \frac{3\ \text{O atoms}}{1\ \text{molecules}} = 2.68827 \times 10^{25}$$

→ **2.7×10^{25} O atoms**

6 Percent composition + two-route check (carbon)

$$\%O = \frac{3(15.999)\text{g/mol}}{84.006\ \text{g/mol}} \times 100\% = 57.135\%$$

Route A: 125 g × 0.57135 = 71.42 g

Route B: $1.4880\ \cancel{\text{mol baking soda}} \times \frac{3\ \cancel{\text{mol O}}}{1\ \cancel{\text{mol baking soda}}} \times \frac{15.999\ \text{g}}{1\ \cancel{\text{mol O}}} = 71.42\ \text{g}$

Both routes → **7.1×10^2 g O** ✓ agree

7 Reverse: from % composition to the formula (per 100g)

Na	27.367	÷	22.990	= 1.1904 mol
H	1.200	÷	1.008	= 1.1905 mol
C	14.298	÷	12.011	= 1.1904 mol
O	57.135	÷	15.999	= 3.5712 mol

÷ 1.1904 → 1.000 : 1.000 : 1.000 : 3.000 → **$NaHCO_3$** (empirical = molecular, n = 1)

8 Reverse: how many boxes contain one mole?

Scale- up: 36 boxes × 125g/box = 4,500g

$$4{,}500\ \text{g} \times \frac{1\ \text{mol}}{84.006\ \text{g}} = 53.5676\ \text{mol}\ NaHCO_3$$

$$1.00\ \text{mol} \times \frac{84.006\ \text{g}}{1\ \text{mol}} \times \frac{1\ \text{box}}{125\text{g}} = 0.672048\ \text{box} \rightarrow \textbf{0.672 box}$$

F. SCALE-UP: FROM ONE BOX TO A WHOLE CLASS

1. ONE BOX	2. ONE CLASS	3. PARTICLES, ONE CLASS	4. REVERSE
125 g of NaHCO3	36 boxes (exact count)	36 boxes	One mole of NaHCO3 is
1.49 mol	**53.6 mol**	**3.23×10^{25}**	**0.672 box**
→ **8.96×10^{24} molecules**	**4,500 g (4.50kg)**	**Formula units of NaHCO3**	**84.006 g, about 67.2% of one box**

G. SIGNIFICANT -FIGURE CHECK

Calculator: $8.960945451 \times 10^{24}$

Reported: 8.96×10^{24} formula units

- **Limit:** The label's 125 g net weight has 3 significant figures.
- **Exact:** Avogadro's constant, chemical subscripts and the count of 36 boxes.
- **Rule:** Keep extra digits during calculations; round only the final answer to 3 significant figures

ONE- NUMBER TAKEAWAY

"A class consist of 36 students, receives 125g box of Queen Pure Baking soda each, would have approximately **53.6 moles** of sodium bicarbonate altogether."

AI use: AI was used as a guide to this card layout, calculation, and references and it were all verified and check by me through following the links, calculating it using a calculator and checking it one by one.

H. REFERENCES (APA 7TH EDITION)

Bureau International des Poids et Mesures. (2019). The International System of Units (SI) (9th ed.). https://www.bipm.org/en/publications/si-brochure/

Commission on Isotopic Abundances and Atomic Weights. (2024). Abridged standard atomic weights 2024. https://www.ciaaw.org/abridged-atomic-weights.htm

Jiang, Z., Cong, Y., Li, X., Xu, J., & Wang, M. (2023). Application of sodium bicarbonate abrasive jet technology for PCB desmear process: Conditions and simulation. Journal of Cleaner Production, 396, Article 136584. https://doi.org/10.1016/j.jclepro.2023.136584

National Center for Biotechnology Information. (n.d.). PubChem compound summary for sodium bicarbonate. https://pubchem.ncbi.nlm.nih.gov/compound/516892